Exercise Sheet 23

Throughout, $G = \mathrm{SL}(2,\mathbb{R})$, $W = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$, $k_\theta = \exp(\theta W)$, $H_0 = \mathrm{diag}(1,-1)$, $S = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $H' = -iW$, $E = \tfrac{1}{2}(H_0 + iS)$ and $F = \tfrac{1}{2}(H_0 - iS)$. Inner products are antilinear in the first slot. Exercise 23.6 is an optional extension.

Exercise 23.1. Hyperbolic coefficients. Let $\gamma = \begin{pmatrix} 1 & i \\ 1 & -i \end{pmatrix}$ and $\varphi(g) = \gamma g \gamma^{-1}$. For $g = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ one has $\varphi(g) = \begin{pmatrix} \alpha & \beta \\ \bar\beta & \bar\alpha \end{pmatrix}$ with $\alpha = \tfrac{1}{2}(a + d + i(c - b))$ and $\beta = \tfrac{1}{2}(a - d + i(b + c))$. For $h = \begin{pmatrix} \alpha & \beta \\ \bar\beta & \bar\alpha \end{pmatrix} \in \mathrm{SU}(1,1)$ put $h \cdot z = (\alpha z + \beta)/(\bar\beta z + \bar\alpha)$ and $j(h,z) = \bar\beta z + \bar\alpha$. For an integer $n \geq 2$, let $\mathcal{A}_n$ be the weighted disk space with orthonormal basis $e_m = c_m^{-1/2} z^m$, where $c_m = \binom{n+m-1}{m}^{-1}$. Let $(\Pi_n(g)f)(z) = j(h^{-1},z)^{-n} f(h^{-1}\cdot z)$ with $h = \varphi(g)$. Put $a_t = \mathrm{diag}(e^t, e^{-t}) = \exp(tH_0)$.

(a) Compute $\varphi(a_t)$.

(b) Show that $(\Pi_n(a_t)1)(z) = (\cosh t - z \sinh t)^{-n}$.

(c) Show that $\langle e_m, \Pi_n(a_t) e_0 \rangle = \binom{n+m-1}{m}^{1/2} (\tanh t)^m (\cosh t)^{-n}$. Check that the squares of these numbers sum to 1.

(d) Differentiate (c) at $t = 0$. Compare with $d\Pi_n(H_0)1$, using $H_0 = E + F$, $d\Pi_n(E) z^m = (n+m) z^{m+1}$ and $d\Pi_n(F) z^m = -m z^{m-1}$.

Exercise 23.2. Real generators as differential operators. Use γ as in Exercise 23.1, with $\gamma^{-1} = \tfrac{1}{2}\begin{pmatrix} 1 & 1 \\ -i & i \end{pmatrix}$. For $X \in \mathfrak{sl}(2,\mathbb{R})$, $d\Pi_n(X)$ is computed from $Y = \gamma X \gamma^{-1} = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$ by

$$d\Pi_n(X) f = n(rz + s) f + \big(rz^2 + (s - p)z - q\big) f'$$

for polynomials f. For $n = 2$ the norms are $\|z^m\|^2 = c_m = 1/(m+1)$, and distinct monomials are orthogonal.

(a) Compute $\gamma H_0 \gamma^{-1}$, $\gamma S \gamma^{-1}$ and $\gamma W \gamma^{-1}$.

(b) Write $d\Pi_n(H_0)$, $d\Pi_n(S)$ and $d\Pi_n(W)$ as differential operators. Deduce the operators for E, F and H'.

(c) For $n = 2$ check that $\langle d\Pi_2(X)1, z \rangle = -\langle 1, d\Pi_2(X) z \rangle$ for $X = H_0$ and $X = S$.

Exercise 23.3. A finite-dimensional comparison module. Let $d \geq 0$ be an integer and $\mathcal{P}_d$ the polynomials of degree at most d. For $h = \begin{pmatrix} \alpha & \beta \\ \bar\beta & \bar\alpha \end{pmatrix} \in \mathrm{SU}(1,1)$ we have $h^{-1} = \begin{pmatrix} \bar\alpha & -\beta \\ -\bar\beta & \alpha \end{pmatrix}$, $h^{-1}\cdot z = (\bar\alpha z - \beta)/(-\bar\beta z + \alpha)$ and $j(h^{-1}, z) = -\bar\beta z + \alpha$. You may use $j(h_1 h_2, z) = j(h_1, h_2 \cdot z)\, j(h_2, z)$ and $(h_1 h_2)\cdot z = h_1 \cdot (h_2 \cdot z)$. Define

$$(\Pi_{-d}(h) f)(z) = j(h^{-1}, z)^d \, f(h^{-1} \cdot z).$$

For $Y = \begin{pmatrix} p & q \\ r & s \end{pmatrix}$, the derivative of Π_{-d} is given by

$$d\Pi_{-d}(Y) f = -d(rz + s) f + (rz^2 + (s - p)z - q) f'.$$

Under conjugation by γ, the elements H', E, F correspond to $\operatorname{diag}(-1,1)$, $\begin{pmatrix}0&0\\1&0\end{pmatrix}$ and $\begin{pmatrix}0&1\\0&0\end{pmatrix}$.

(a) Show that $\Pi_{-d}(h)$ maps $\mathcal{P}_d$ to itself and that $\Pi_{-d}(h_1h_2) = \Pi_{-d}(h_1)\Pi_{-d}(h_2)$.

(b) Compute H', E and F on z^m, and list the K-types.

(c) For $d = 2$ write the matrices in the basis $(1, z, z^2)$. Verify $[E,F] = H'$ and $H'^2 + 2H' + 4FE = 8I$.

(d) Show that for $d \geq 1$ no positive definite inner product on $\mathcal{P}_d$ satisfies $\langle Eu, v\rangle = -\langle u, Fv\rangle$ for all u, v.

Exercise 23.4. Lowest-weight modules and descent. Let $\lambda \in \mathbb{R}$. Let M have basis $u_0, u_1, \ldots$ with

$$H'u_m = (\lambda + 2m)u_m, \qquad Eu_m = u_{m+1}, \qquad Fu_0 = 0, \qquad Fu_m = -m(\lambda + m - 1)u_{m-1}.$$

(a) Verify $[E,F] = H'$ on each u_m. Show that $\Omega = H'^2 + 2H' + 4FE$ acts by $\lambda(\lambda - 2)$.

(b) Suppose M has a Hermitian form with $\langle u_0, u_0\rangle = 1$, $\langle u_j, u_m\rangle = 0$ for $j \neq m$, and $\langle Ex, y\rangle = -\langle x, Fy\rangle$. Show that $\langle u_m, u_m\rangle = m!\,\lambda(\lambda+1)\cdots(\lambda+m-1)$.

(c) Deduce that the form is positive definite if and only if $\lambda > 0$.

(d) Let $n \geq 2$ be an integer and $(n)_m = n(n+1)\cdots(n+m-1)$. On $\mathcal{A}_n$, $Ez^m = (n+m)z^{m+1}$, $Fz^m = -mz^{m-1}$, $H'z^m = (n+2m)z^m$ and $\|z^m\|^2 = m!(n-1)!/(n+m-1)!$. Show that $u_m \mapsto (n)_m z^m$ intertwines the actions and matches the norms of (b) when $\lambda = n$.

(e) Suppose M is the K-finite part of a representation of G, with u_m of K-type $\lambda + 2m$, and the form is positive. Use $k_{2\pi} = I$ and $k_\pi = -I$ to show that λ is a positive integer, and an even one if the representation factors through $\mathrm{PSL}(2,\mathbb{R})$.

Exercise 23.5. Even modules and the complementary series. Let $c \in \mathbb{R}$. Let V have basis v_k, $k \in 2\mathbb{Z}$, with

$$H'v_k = kv_k, \qquad Ev_k = v_{k+2}, \qquad Fv_k = \tfrac{1}{4}\big(c - (k-2)k\big)v_{k-2}.$$

(a) Verify $[E,F] = H'$ and that $\Omega = H'^2 + 2H' + 4FE$ acts by c.

(b) Suppose V has a Hermitian form with orthogonal v_k and $\langle Ex, y\rangle = -\langle x, Fy\rangle$. Show that $\|v_{k+2}\|^2 = \frac{1}{4}(k(k+2) - c)\|v_k\|^2$. Deduce that a positive definite such form exists if and only if $c < 0$.

(c) For $c = -\frac{3}{4}$ and $\|v_0\| = 1$, compute $\|v_2\|^2$, $\|v_4\|^2$ and $\|v_{-2}\|^2$.

(d) For $c = 0$, find invariant subspaces with K-types $\{k \geq 2\}$ and $\{k \geq 0\}$. Identify the K-types of the pieces.

Exercise 23.6. Optional extension: square integrability. Keep the notation of Exercise 23.1. Take as given the following facts about G. Every $g \in G$ can be written $g = k_\theta a_t k_\psi$ with $t \geq 0$. There is a constant $C > 0$ such that every continuous $\Phi \geq 0$ on G with $\Phi(k_\theta g k_\psi) = \Phi(g)$ satisfies

$$\int_G \Phi(g)\, dg = C \int_0^\infty \Phi(a_t) \sinh(2t)\, dt$$

for Haar measure dg. Also $\Pi_n(k_\theta)e_0 = e^{in\theta}e_0$.

(a) Show that $|\langle e_0, \Pi_n(g)e_0\rangle| = (\cosh t)^{-n}$ for $g = k_\theta a_t k_\psi$.

(b) For real s, show that $I(s) = \int_0^\infty (\cosh t)^{-2s} \sinh(2t)\, dt$ is finite if and only if $s > 1$, with $I(s) = 1/(s-1)$.

(c) Conclude that this coefficient is square integrable for $n \geq 2$. Explain why the same function with $n = 1$ is not.